

SANTA CRUZ DE LA SIERRA, Bolivia

WMO: 852440

Lat: 17.645S

Lon: 63.135W

Elev: 373

StdP: 96.92

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.8	11.1	0.0	3.9	15.3	2.1	4.6	16.2	13.9	24.5	13.0	24.2	5.7	170	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.6	35.1	23.5	34.1	23.7	33.2	23.9	26.1	31.0	25.7	30.5	25.5	30.2	7.6	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	21.1	28.1	24.4	20.2	27.3	24.2	20.0	27.1	83.5	31.3	81.9	30.9	80.4	30.3	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	11.4	10.2	6.9	37.1	1.7	1.3	5.7	38.0	4.7	38.7	3.7	39.4	2.5	40.3		
(5)			4.8	27.4	1.8	0.7	3.6	27.9	2.5	28.3	1.5	28.7	0.2	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.3	26.5	26.1	25.7	24.4	21.7	20.6	20.7	22.8	24.8	26.0	26.2	26.3
	DBStd	3.71	1.69	1.75	1.90	2.85	3.53	3.91	4.08	3.84	3.72	2.85	2.39	1.90
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD18.3	79	0	0	0	2	11	24	27	11	4	1	1	0
	CDD10.0	5216	512	450	488	431	364	317	331	396	443	495	484	504
	CDD18.3	2252	254	217	230	182	117	91	98	147	198	237	235	246
	CDH23.3	18722	2131	1647	1727	1337	717	524	709	1340	1972	2282	2239	2098
	CDH26.7	7440	847	601	633	483	203	122	213	574	931	1037	976	820
Wind	WSAvg	5.0	4.6	4.5	4.2	4.5	4.8	5.4	5.6	5.9	5.7	5.4	5.1	4.9
Precipitation	PrecAvg	1399	182	185	135	110	94	76	55	33	77	102	136	186
	PrecMax	2075	365	356	318	324	279	196	301	100	194	267	248	342
	PrecMin	939	92	78	76	29	10	0	2	0	1	21	34	59
	PrecStd	264	71	60	57	67	69	52	60	30	53	63	66	79
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	33.9	33.8	33.1	31.4	30.2	31.8	34.1	36.0	36.5	35.8	34.9
		MCWB	23.9	25.0	25.0	24.8	23.9	23.2	21.8	22.5	22.7	23.3	23.7	24.2
	2%	DB	33.8	32.8	32.2	32.0	30.2	29.2	30.1	33.0	34.9	34.9	34.1	33.2
		MCWB	24.4	24.8	24.6	24.6	23.6	22.8	21.9	22.1	22.9	23.7	23.8	24.6
	5%	DB	32.2	31.8	31.2	31.0	29.1	28.1	29.1	31.9	33.8	33.3	33.0	32.1
		MCWB	24.7	24.7	24.4	24.3	23.2	22.4	21.6	22.1	23.0	23.7	23.9	24.6
	10%	DB	31.0	30.2	30.2	29.9	27.7	26.9	27.9	30.2	32.1	32.1	31.9	31.0
		MCWB	24.4	24.5	24.3	24.1	22.5	21.8	21.2	21.7	22.6	23.6	23.8	24.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.6	26.2	25.9	25.3	24.7	23.9	24.5	25.3	25.8	26.1	26.4
		MCDB	31.6	31.2	30.8	30.6	29.2	28.2	28.9	30.8	32.1	31.8	31.8	31.5
	2%	WB	25.9	25.8	25.6	25.5	24.6	23.8	23.2	23.6	24.6	25.2	25.6	25.7
		MCDB	30.5	30.1	30.1	30.0	28.5	27.6	27.8	30.0	31.4	31.3	31.2	30.6
	5%	WB	25.5	25.4	25.3	25.0	24.0	23.2	22.6	23.1	24.0	24.8	25.1	25.4
		MCDB	29.9	29.6	29.6	29.6	27.6	26.5	26.9	29.2	30.8	30.8	30.6	30.0
	10%	WB	25.0	25.0	24.9	24.5	23.4	22.6	21.9	22.4	23.3	24.3	24.7	25.0
		MCDB	29.3	28.9	28.9	28.5	26.3	25.5	25.9	28.0	29.9	29.9	29.7	29.4
Mean Daily Temperature Range	5% DB	MDBR	8.6	8.2	8.6	9.1	8.5	8.6	10.1	11.6	11.4	10.3	9.9	8.7
		MCDBR	10.9	10.3	10.1	10.6	10.3	10.0	11.8	13.5	13.3	12.2	12.1	10.7
		MCWBR	3.9	3.8	3.9	4.3	4.5	4.7	5.0	5.1	4.6	4.1	4.4	3.8
	5% WB	MCDBR	9.0	8.7	8.8	9.3	8.6	8.6	10.1	11.6	11.6	10.3	9.8	9.0
		MCWBR	3.7	3.8	3.7	4.1	4.2	4.4	4.8	5.2	4.6	4.1	3.9	3.7
Clear-Sky Solar Irradiance	taub		0.402	0.397	0.387	0.375	0.372	0.370	0.386	0.504	0.631	0.522	0.427	0.410
	taud		2.435	2.463	2.488	2.497	2.457	2.488	2.397	2.012	1.713	2.033	2.317	2.392
	Ebn at Noon		943	938	925	899	864	847	841	765	706	817	914	935
	Edh at Noon		124	119	112	104	101	95	106	166	239	180	138	129
All-Sky Solar Radiation	RadAvg		5.59	5.21	5.04	4.54	3.65	3.35	3.77	4.43	4.83	5.25	5.58	5.44
	RadStd		0.61	0.39	0.37	0.34	0.49	0.34	0.32	0.32	0.47	0.42	0.43	0.42

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.29	N/A	N/A	+0.58	N/A	N/A	N/A	+104	+94
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page